

Problem

Construct the Bode magnitude and phase plots for

$$H(s) = \frac{2(s+1)}{(s+2)(s+10)}, \quad s = j\omega$$

Step-by-step solution

Step 1 of 7

Consider the following transfer function:

$$H(s) = \frac{2(s+1)}{(s+2)(s+10)}$$

Substitute $j\omega$ or s .

$$\begin{aligned} \mathbf{H}(\omega) &= \frac{2(j\omega+1)}{(j\omega+2)(j\omega+10)} \\ &= \frac{2(j\omega+1)}{20\left(1+\frac{j\omega}{2}\right)\left(1+\frac{j\omega}{10}\right)} \\ &= \frac{(j\omega+1)}{10\left(1+\frac{j\omega}{2}\right)\left(1+\frac{j\omega}{10}\right)} \\ &= \frac{0.1(j\omega+1)}{\left(1+\frac{j\omega}{2}\right)\left(1+\frac{j\omega}{10}\right)} \dots\dots (1) \end{aligned}$$

Step 2 of 7

Write the magnitude expression from equation (1).

$$\begin{aligned} G_{dB} &= 20\log_{10}(0.1) + 20\log_{10}|1+j\omega| - 20\log_{10}\left|1+\frac{j\omega}{2}\right| - 20\log_{10}\left|1+\frac{j\omega}{10}\right| \\ &= -20\log_{10}(10) + 20\log_{10}|1+j\omega| - 20\log_{10}\left|1+\frac{j\omega}{2}\right| - 20\log_{10}\left|1+\frac{j\omega}{10}\right| \\ &= (-20)(1) + 20\log_{10}|1+j\omega| - 20\log_{10}\left|1+\frac{j\omega}{2}\right| - 20\log_{10}\left|1+\frac{j\omega}{10}\right| \\ &= -20 + 20\log_{10}|1+j\omega| - 20\log_{10}\left|1+\frac{j\omega}{2}\right| - 20\log_{10}\left|1+\frac{j\omega}{10}\right| \dots\dots (2) \end{aligned}$$

Step 3 of 7

There are three corner frequencies at $\omega = 1, 2, 10 \text{ rad/s}$.

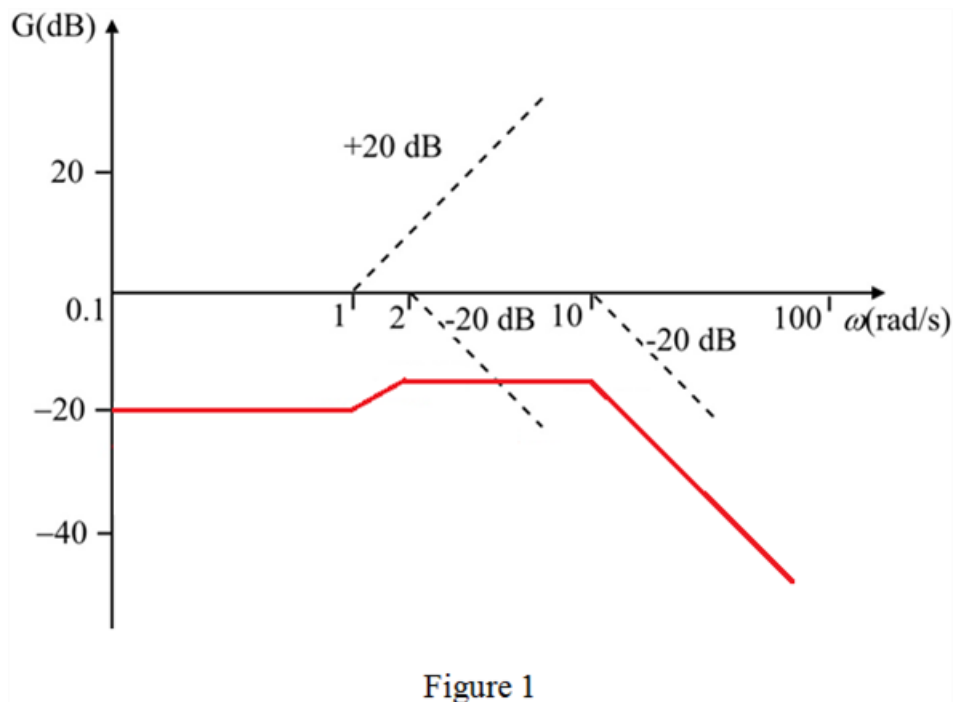
For the zero with corner frequency at $\omega = 1 \text{ rad/s}$, the slope of the magnitude plot is 20 dB/decade .

For the pole with corner frequency at $\omega = 2 \text{ rad/s}$, the slope of the magnitude plot is -20 dB/decade .

For the pole with corner frequency at $\omega = 10 \text{ rad/s}$, the slope of the magnitude plot is -20 dB/decade .

Step 4 of 7

Draw the asymptotic magnitude plot.



Step 5 of 7

Now write the phase expression from equation (1).

$$\phi = \tan^{-1}(\omega) - \tan^{-1}\left(\frac{\omega}{2}\right) - \tan^{-1}\left(\frac{\omega}{10}\right) \dots (3)$$

Table 1

$\omega(\text{rad/s})$	0.01	0.1	1	10	100	1000
ϕ	0°	2.27°	12.72°	-39.4°	-83.72°	-89.37°

Step 6 of 7

Draw the asymptotic phase plot.

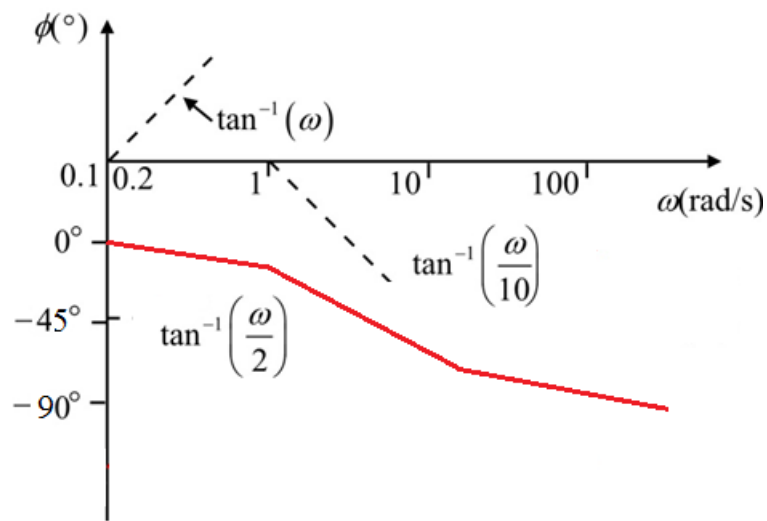


Figure 2

Step 7 of 7

The Bode diagram program is,

```
num=[2 2];
den=[1 12 20];
sys=tf(num,den);
bode(sys)
grid
```

MATLAB output for Bode plot is shown in Figure 3.

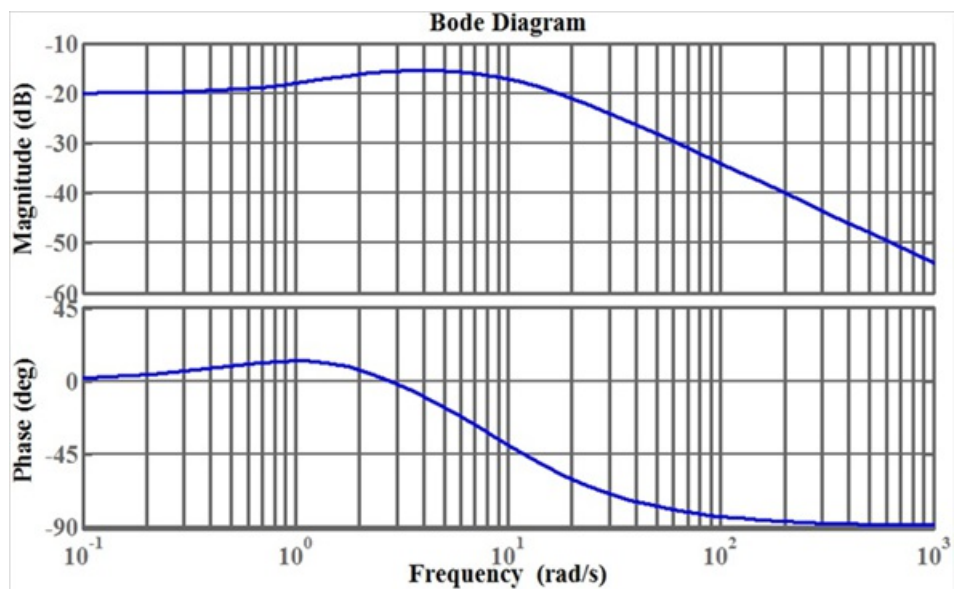


Figure 3

Thus, the Bode plots are drawn.